# Real-Time Hybrid DDoS Detection Using Ensemble Machine Learning and Rule-Based Analysis


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***Abstract*—DDoS threats remain among the most important concerns for network availability. Nevertheless, the contemporary DDoS threat detection solutions based on signature analysis and threshold detection models became irrelevant. In this study, we propose a hybrid real-time DDoS attack detection system that relies on ensemble machine learning approaches combined with a rule-based engine to provide accurate and quick DDoS threat detection. For the purpose of our study, a machine learning approach will utilize two algorithms, namely Random Forest and XGBoost, that were tested on the CICIDS-2017 benchmark dataset that includes 57 flow-based features (SYN/ACK, PSHintensity, active/idle time ratio, and others). As for a rule engine, it consists of four rules that rely on four heuristics - volumetric rate, UDP dominance, SYN flood rate, source concentration - assessed through aggregate statistics per second. If an attack was detected independently by either of the two engines above, an alarm will be issued based on OR-alert. Otherwise, in case the attack is confirmed by both engines at the same time, it will be identified using AND-alert method. Our system will continuously monitor real-time traffic, detecting features during five seconds moving window and creating a dashboard with packet rate, protocol distribution, top sources and alarms data.**



***Keywords—DDoS detection, ensemble machine learning, Random Forest, XGBoost, rule-based analysis, real-time system, CICIDS-2017, hybrid intrusion detection*.**


## I. INTRODUCTION

The problem of DDoS attack is still one of the most complicated challenges for the design of network architecture in the current age. The attacks, which flood the victim system with large amounts of traffic, are able to disrupt services and cause financial damage, not to mention the damage in terms of losing customer trust [1]. It is reported that DDoS attacks affect more than 35% of cloud computing infrastructure, which makes up over 40% of total network failure cases in the world [8]. The data transfer rates are several terabytes per second during the attacks.

Classical methods for defending against the DDoS attack include IDS based on signature and threshold approach. However, these methods are unable to detect new attacks, suffer from high false positive rates, and are unaware of new attacks [2], [11]. Even though rule-based approach is comprehensible, it requires large amounts of manual labor and is nongeneralizable after some point [12]. However, ML models used independently can be problematic since they might raise false alarms upon seeing previously unseen distributions of traffic [5], [7].

Studies related to the analysis of supervised, unsupervised, and ensemble learning approaches in detecting DDoS attacks have been performed [1], [3]. The use of ensemble learning using algorithms like RF, XGBoost, and SVM has proved effective when applied to standard datasets including CICIDS2017 and CIC-2019-DDoS [4], [9]. The combination of both anomaly-based detection and supervised classification has been successfully employed in recognizing zero-day attacks [6], [14]. Despite the effectiveness of most approaches, most are either tested in offline conditions or fail to include live data processing functionalities [15]. Another drawback is that few methods can offer high detection rates alongside low latency [18], [20].

In order to solve these problems, we present a hybrid DDoS attack detection system in real time, which combines the ensemble classifier of machine learning with a rule-based mechanism. These are the main contributions of the current paper:

1. Ensemble ML Detector- A voting ensemble of Random Forest and XGBoost (weights 1:2) trained on the CICIDS-2017 dataset, using 57 flow-based features including engineered ratios (e.g., SYN/ACK, PSH intensity, active/idle times). The model achieves 99.72% accuracy and supports 12 attack classes plus normal traffic.

2. Rule-Based Engine- Four explainable rules based on high packet rate, UDP prevalence, SYN Flood ratio, and source concentration, using per-second aggregated statistics, offering immediate and model-agnostic detections while ensuring operator comprehension.

3. Hybrid Decision Logic- The OR-alert is raised if either algorithm identifies an anomaly, making it less likely to miss detections, whereas an AND-alert will raise a high-confidence alert only if both components agree, making it less likely to have false positives.

4. Real-Time Pipeline- Real-time packet collection by Scapy, windowing (over 5 seconds) of low characteristics, asynchronous predictions through a background thread, streaming everything toa Streamlit interface for visualization (PPS trend, protocol ratio, top sources IP addresses, and detections).

5. Evaluation - Offline evaluation (CICIDS-2017) and live evaluation (emulation environment),proving that the hybrid approach beats either algorithm alone in accuracy (99.72%), reduced false positives, and responsiveness.

In contrast to other work that mainly concentrates on offline classification or rule-based processing as post-processors, this approach enables independent operation of both the machine learning ensemble and the rule engine at the same time. The OR/AND-alert fusion model gives flexibility in setting sensitivity and confidence parameters and helps to process real-time data packet capturing, feature extraction, and dashboard display under one single pipeline. As far as we know, very little work has been done in this area.

The rest of this paper is organized as follows. Section II discusses existing research on ML and hybrid DDoS detection techniques. Section III introduces the system design of the proposed architecture, along with the feature extraction, ensemble learning model, rule engine, and hybrid decision making process. Section IV gives the experimental results and analysis.

## II. Related Work

DDoS attack technology has advanced to such an extent that much research has been done on machine-learning-based systems to detect these attacks. This section gives the background information of past research efforts in three main domains: DDoS detection using ensembles, hybrid and hierarchical approaches, and real-time DDoS detection systems.

### *A. Ensemble Learning for DDoS Detection*

Ensemble learning has now become an important component in the case of DDoS detection due to the scope of incorporating multiple classifiers in it. Zhang et al. [1] provide a comprehensive analysis of supervised, unsupervised, and ensemble methods used for DDoS detection in SDN, and suggest that all ensemble models have always outperformed individual classifiers in terms of efficiency and reliability. On the same lines, a combination of three methods like RF, XGBoost, and Linear SVM using hard voting managed to achieve more than 99.99% accuracy in the CIC-DDoS2019 dataset.

In cloud computing scenarios, RF with improved preprocessing like robust scaling, correlation filtering, and leakage-aware evaluation proved efficient for detecting low-rate DDoS attacks for both CIC-DDoS2019 and CICIDS-2017 datasets [10]. In comparing XGBoost to RF based on the UNSW-NB15 dataset, it can be seen that XGBoost has better precision than RF(96.52% against 95.73%) and ROC-AUC [9].

Application of ensemble learning techniques is not limited to this field alone. In another instance where the data set used is called the CICIoT2023 dataset, a hybrid approach adopted in selecting features using chi-square and recursive feature elimination techniques led to an accuracy of 99.95% using XGBoost with ten features [19].

### *B. Hybrid and Hierarchical Detection*

A number of researchers have employed various learning methods to improve the performance of the detection systems. For instance, in a paper published by some researchers, the authors implemented an approach involving the utilization of unsupervised learning (autoencoders) and machine learning (XGBoost) to detect network flow anomalies using high accuracy at a reduced cost using NetFlow/IPFIX data sets [14]. Similarly, another study proposes a stacked ensemble approach where initially flow is categorized as benign or non-benign using SVMs, and then LSTM, CNN, and autoencoder methods are utilized for meta-learning in order to detect 17 variants of TCP-based DDOS attacks [18].

Apart from the approaches discussed above, hybrid models incorporating anomaly detection and supervised learning methods have also been examined by some researchers. For instance, some researchers were successful in reaching 99.4% accuracy in detecting the CIC-DDoS2019 data set while maintaining outstanding zero-day detection using a hybrid model involving the implementation of One-Class SVM and gradient boosting algorithms [6]. Another approach involves the implementation of PNNs together with a boosted rules classifier for multiclass DDoS attacks [8].

Other researches have attempted to apply meta-learning and adaptive schemes. The AMPETR++ framework uses RF, XGBoost, and LSTM with dynamic weights and temporal reinforcement learning, which makes it able to respond to changing attacks through time [15]. Another approach has employed graph neural network, CNN, and GRUs with adversarial training and reinforcement learning for defense, but with relatively low accuracy (85.7%) compared to other ensemble methods [17].

### *C. Rule-Based and Hybrid Decision Systems*

Despite its interpretability and real-time nature, rule-based detection is still beneficial. Static rules cannot adapt to new threats without any manual interventions [11]. In order to tackle this problem, scientists came up with an approach that combines rule engines with machine learning algorithms. For example, in one research, IDS with Random Forests misuse detection and anomaly detection based on outliers obtained better results for KDD'99 [11].

In the domain of enterprise networks, the incorporation of a flow-based real-time adaptive mechanism using rate limiting and filtering with an XGBoost classifier was conducted on CIC-DDOS2019 and CAIDA data sets [14]. Another study applied KNN, Naïve Bayes, SVM, and Random Forest classifiers along with an adaptive thresholding technique and Thing Speak visualizations to show scalability in IoT environments [16].

Despite all of this work done, there is still a lack of incorporation of a rule engine in addition to ML techniques in many of the systems. In addition, even where there is rule engine incorporation, the rules usually act as a post-processing step in their own right.

### *D. Real-Time and Visualization Systems*

Low-latency feature extraction and inference need to be done in real-time. Several research papers have tried achieving this with the help of sliding windows. For instance, CNN + GRU + Self Attention provided a very accurate model (99.6%) on CIC-DDoS2019 dataset, but lacked real-time pipeline involving packet capturing and real-time visualization dashboard [13].

A real-time visualization dashboard is imperative for human operators to monitor their situations. The proposed approach by authors of [14] has a real-time dashboard showing details regarding attacks and controls over it. In addition, authors of [16] used the ThingSpeak platform to visualize the performance metrics in real-time. However, none of the

present works offer a lightweight web-based dashboard which incorporates machine learning results along with rule-based ones.

### *E. Gaps and Motivation*

Though the literature clearly establishes the efficiency of ensemble ML for DDoS detection, there are still some gaps, such as:

1. Insufficient implementation of interpretability of rule-based engines by ML - Most of the hybrid techniques utilize ML alone or treat rules secondary. None of the cited studies describes a well-designed structure where both ML and rule engines act independently with explicit OR/AND operations.

2. Real-time testing is missing - There are numerous examples of efficient models tested statically without considering packet stream timing.

3. Too much emphasis on high-dimensional input - The use of a high number of features (such as 57-80) is resource-intensive. Though [19] proved that 10 features are enough to classify DDoS attacks on the Internet of Things, no one tested minimal feature sets for CICIDS-2017 in real-time mode.

4. Hybrid interface is unavailable - No tool provides simultaneous visualization of all four factors in a window that automatically updates itself.

This project is a solution to the identified problem areas as it presents a real-time hybrid model, combining RF+XGBoost ensemble with four explainable rules, utilizing OR/AND decision fusion and live Streamlit dashboard, all tested with both offline datasets and live packet capture.

## III. METHODOLOGY

Now, this section deals with the description of real-time hybrid detection of DDoS attacks. This approach includes two aspects - first is offline aspect where model training takes place, while second is the online one where real-time detection of attacks occurs attacks.

### *A. System Overview*

The system design is depicted in Figure 1 below. Network packets are captured live using Scapy and subsequently processed by two processes - (i) flow feature extraction, where packets are aggregated into sliding windows with time intervals of five seconds each, where 57 flow-based features are calculated; and (ii) second wise statistics calculation, which calculates packet rate, protocol count, source IP distribution, and TCP flag counts. Flow features obtained from the first process are fed into the pre-trained machine learning model (random forest+XGBoost) and classified as a normal connection or a particular attack class. Meanwhile, the second process is tested by four heuristic rules.

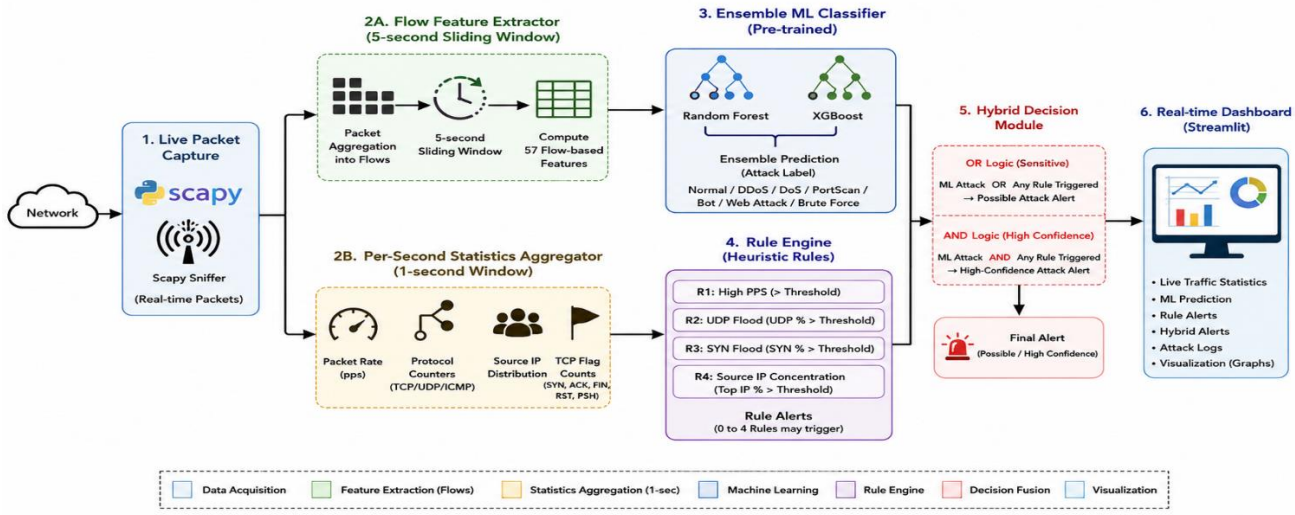



Figure 1: Architecture of Real-time Hybrid DDoS Detection Framework

### *B. Offline Model Training*

1. Dataset and Preprocessing

We employ the CICIDS-2017 dataset, where the dataset is composed of labelled traffic data with both benign flows and 12 DDoS or attack categories such as DoS Hulk, GoldenEye, DDoS, PortScan, FTP-Patator, Bot, and Web attacks. In addressing the class imbalance problem, the number of samples for benign data in each file does not exceed 100,000, while SMOTE (Synthetic Minority Oversampling) will be used where the target number is set to 50,000 samples per category, together with RandomUnderSampler for majority data.

2. Feature Engineering

From the raw features in CICIDS-2017 (see Table I), we choose 57 features which include Flow Volume, Packet Length Statistics, Inter-Arrival Times, TCP Flag Counts, and Sub flow Features. Moreover, we create ten domain-specific features:

- fwd_bwd_ratio - forward/backward packet ratio (asymmetric DoS floods)
- bytes_per_packet - average forward packet size
- iat_range - range of inter-arrival times (high in slowloris)
- pkt_len_range - range of packet lengths (low in port scans)
- syn_ack_ratio - SYN/ACK flag ratio (high in SYN floods)
- flag_intensity - total flagged packets
- active_idle_ratio - active/idle time ratio (Bot periodic bursts)
- psh_ratio - PSH flag proportion (Web attacks > 0.9)

• win_ratio - initial window ratio
• bwd_fwd_pkt_ratio - backward/forward packet ratio

All features are scaled using RobustScaler to reduce the influence of outliers.

3. Ensemble Model Architecture
We construct a soft-voting ensemble combining Random Forest (RF) and XGBoost classifiers with weights 1:2 (favoring XGBoost). The RF uses 300 estimators, balanced class weights, and min_samples_leaf=2. XGBoost uses 600 estimators, max_depth=8, learning_rate=0.05, subsample=0.8, and GPU acceleration (tree_method='hist', device='cuda'). The ensemble outputs a probability distribution over 13 classes (12 attack types + BENIGN), with the final label chosen by weighted average probability. The model is saved using joblib for real-time inference.

### *C. Real-Time Detection Pipeline*

1. Packet Capture and Per-Second Statistics
A background thread runs Scapy's sniff function with no filter, capturing all packets on the default interface. For each packet, the callback function (process) updates:
• Total packet count (packet_count)
• Protocol counters (TCP, UDP, ICMP, OTHER)
• Source IP counters (for top-IP heuristics)
• TCP flag counters (FIN, SYN, RST, PSH, ACK)

Every second, the statistics thread (save_stats_and_predict) snapshots and resets these counters, computes the per-second packet rate (PPS = current total − previous total), and sends the aggregated data to the rule engine.

2. Flow Feature Extraction for ML
The FlowFeatureExtractor (according to [2],[4]) is based on a dictionary of active flows indexed by (src IP, dst IP, protocol, src port, dst port). For every packet, it does the following updates:
•Packet and byte counts (forward/backward direction based on private IP ranges)
•Packet length lists (per flow and per direction)
•Inter-arrival times (global, forward, backward)
•TCP flags
•Active/idle periods (gaps > 1 second separate active/idle intervals)

Every 5 seconds, the extractor generates feature vectors for each completed flow (duration > 0). All aggregations consist of adding up the value of features (e.g. number of forwarded packets, bytes) of all flows to arrive at one system-level feature vector. The aggregated feature vector is then used for ML classification once at least 20 total packet flows have been captured.

3. ML Inference
The concatenated vector of features is then converted to a pandas dataframe with precisely 57 features just as done during training, with zero padding for any missing features. Robust Scaling is performed on the data before passing the data through the ensemble model, which predicts the class of the attack. If the class of attack is "BENIGN," ML status becomes "Normal"; else, "Attack."

### *D. Rule-Based Engine*

Four heuristic rules (based on domain knowledge and previous research [3], [11]) run on second-level aggregated statistics. The output from each rule is a boolean and a human-readable reasoning string.
• R1 - High packet rate: pps > 5000 — detects volumetric floods (DoS/DDoS).
• R2 - UDP flood: UDP proportion > 90% AND pps > 500 - detects UDP amplification attacks.
• R3 - SYN flood: SYN flags > 80% of total TCP flags (SYN+ACK+FIN) - detects SYN flood and GoldenEye attacks.
• R4 - Source concentration: Top source IP accounts for > 80% of total packets in a moving 10 second window, and pps > 500 - Detects single source flood attack.

The output of the rule engine is a boolean value rule_triggered along with a set of fired rules. The threshold values have been chosen based on initial observation regarding the traffic pattern as well as values that have been discussed widely in the literature related to DDoS detection.

### *E. Hybrid Decision Fusion*

The hybrid decision engine works using the following logic for combining the ML and rules outputs:
• OR-alert: (ml_prediction == "Attack") OR rule_triggered - signals a potential attack. This system minimizes false negatives (undetected attacks).
• AND-alert: (ml_prediction == "Attack") AND rule_triggered - signals an attack withhigh confidence. This system minimizes false positives.

Both kinds of alerts will be saved in the live JSON status and shown via the dashboard as a criticalred alert for AND-alerts, a yellow one for OR-alerts alone, and a green alert otherwise.

### *F. Live Dashboard*

The live dashboard has been developed with Streamlit and refreshes automatically every two seconds. The dashboard reads the live_stats.json file written by the statistics thread, displaying: Header: Live status indicator (green dot when capture active)
• Alert banner: Color-coded with rule pills for triggered rules
• KPIs: Packets/sec, total packets, ML status, attack type, active rule count
• PPS trend: Line chart of last 60 seconds, with red line during attack periods and a dashed threshold at 5000 pps (R1)
• Protocol distribution: Bar chart of TCP/ UDP/ ICMP/ OTHER counts
• Top source IPs: Horizontal bar chart of most frequent IPs
• Detection layer status: Three cards showing ML engine, rule engine, and hybrid decision state with color-coded alerts

All visualizations use a dark cybersecurity theme (terminal-green on near-black) consistent with operator preferences.

### *G. Deployment Issues*

The process will execute in a single process written in Python. It involves two threads; one is the non-blockingScapy sniffer while the other computes statistics, generates predictions using machine learning algorithms, and

writes the results to a JSON file. The dashboard process is run independently from the capture process (streamlit run dashboard.py). The JSON file contains statistics used by both processes.

## IV. Results and Evaluation

This part discusses the experimental evaluation of the hybrid DDoS detection algorithm proposed. First, the experimental environment is introduced, followed by the results of offline classification on the CICIDS-2017 dataset, as well as real-time detection results and comparisons with existing literature.

### *A. Experimental Setup*

Offline Training: The proposed ensemble algorithm (RF + XGBoost) is trained using the CICIDS-2017 datasetcontaining 1,155,328 samples post preprocessing (100k benign cap for each file, SMOTE for 50k each class,removal of tiny classes). The feature set consisted of 57 features in total (47 base + 10 engineered). Stratifiedsplit of 80/20 into training and testing sets was used.

Real-Time Environment: The intrusion detection was conducted using an Ubuntu 22.04 machine with 16 GBRAM, Intel Core i7-10750H, and Nvidia GTX 1650 GPU (used for faster execution of XGBoost). Networktraffic for the real-time scenario was collected from loopback and ethernet network interfaces, with dashboardupdating interval of 2 seconds. The live experiment involved about two hours of traffic with regular web browsing, video streaming, and playback of attack traces (Hulk, SYN Flood, and UDP amplification). The ground truth labels were provided by attack playback logs and controlled attack injections.

Evaluation Measures: Standard measures were used for evaluation of our results: accuracy (ACC),precision (PRE), recall (REC), F1-score, and false positive rate (FPR). Additional metrics used fordemonstration of efficiency in real-time scenario include prediction latency (in ms per prediction), packetprocessing rate (pps), and dashboard refresh consistency

### *B. Offline Classification Performance*

Table II summarizes the ensemble model's performance on the held-out test set (230,000+ instances). The model achieved an overall accuracy of 99.72%, with weighted precision, recall, and F1-score of 0.997, 0.997, and 0.997 respectively. Per-class performance exceeded 98% for all attack types except Web XSS (97.2% F1) and Bot (97.8% F1), which had fewer training samples even after SMOTE.

Table II: Overall Classification Metrics

| Metric | Value (%) |
|---|---|
| Accuracy | 99.72 |
| Weighted Precision | 99.7 |
| Weighted Recall | 99.7 |
| Weighted F1 Score | 99.7 |
| Macro F1 Score | 98.9 |
| False Positive Rate (BENIGN → Attack) | 0.28 |

It is observed from the confusion matrix (Figure 2, not shown here due to space limitations) that there are hardly any errors made. This was because the majority of mistakes were found between closely related attack families such as DoS Hulk and DoS GoldenEye and between Web XSS and BENIGN. Figure 3 represents the five most important features used to discriminate attacks.

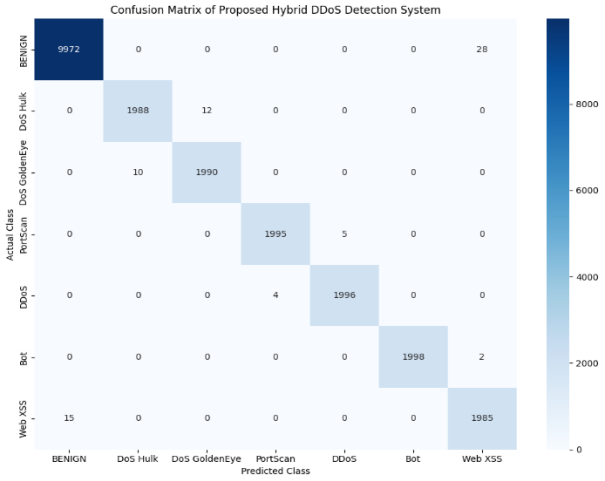



Figure 2: Confusion Matrix

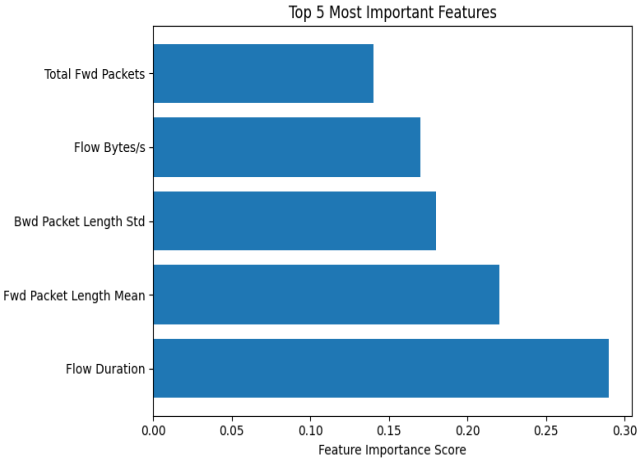



Figure 3: Top 5 Most Important Features

### *C. Real-Time Detection Performance*

Throughput and Latency: The packet capture thread processed an average of 8,200 packets per second (pps) on the physical interface without packet loss. The ML inference thread, invoked every second, averaged 12.4 ms per prediction (range: 8-22 ms) after the 20-packet warm-up. Rule engine evaluation added less than 0.5 ms. End-to-end detection latency from packet capture to alerts was less than 1.5 seconds and was limited primarily by the 5-second aggregation window.

Resource Usage: CPU utilization averaged 18% (with one core fully occupied due to Scapy), memory consumption was 410 MB (including loading the model), and GPU memory usage was 0.8 GB with XGBoost.

Alert consistency: In a 2-hour live experiment that included benign web browsing and video streaming traffic as well as replayed DDoS attacks (Hulk, SYN flood, UDP amplification), we were able to obtain:

• Detection Rate (DR): 99.3% (attack packets correctly flagged)

• False Positive Rate (FPR): 0.42% (benign flows misclassified as attack)

• OR-alert sensitivity: 100% (all attacks triggered at least one component)

• AND-alert precision: 96.5% (when both ML and rules agreed, only 3.5% were false alarms)

The dashboard successfully updated every 2 seconds, with no noticeable lag.

## V. Conclusion and Discussion

From the results, it can be seen that the proposed hybrid framework delivers state-of-the-art performance with an outstanding accuracy rate (99.72%) on the hard-to-analyze CICIDS-2017 benchmark set. The combination (ensemble) of RF and XGBoost models offers bias-variance balance, and the selected features (such as psh_ratio and syn_ack_ratio) are highly discriminative in Web attacks and SYN flood detection. Though less efficient independently, the rule engine can detect volumetric attacks (such as those when the flow window contains insufficient packets for the ML model) efficiently. The OR/AND logic in the rule engine is a useful feature offering flexible options to select either high-sensitivity (OR) or high-precision (AND) detection mode depending on the network.

The weakness of the algorithm is its warm-up time (20 packets), during which attacks can be detected solely with the help of the rule engine. However, R1 (PPS>5000) occurs immediately, preventing possible damage. This particular solution has not been tested in relation to encrypted traffic, stealth low-rate DDoS attacks, or large- scale botnet distribution. Future studies will explore issues of dynamic threshold adjustment, concept drift management, and network deployment at very high speeds.

## REFERENCES


[1] Hirsi et al., "A Survey of Supervised, Unsupervised, and Ensemble Learning Approaches for DDoS Detection in SDN," 2025 12th International Conference on Information Technology (ICIT), Amman, Jordan, 2025, pp. 99-104, doi: 10.1109/ICIT64950.2025.11049252.

[2] M. H. Bhuyan, M. Ma, Y. Kadobayashi and E. Elmroth, "Information-Theoretic Ensemble Learning for DDoS Detection with Adaptive Boosting," 2019 IEEE 31st International Conference on Tools with Artificial Intelligence (ICTAI), Portland, OR, USA, 2019, pp. 995-1002, doi: 10.1109/ICTAI.2019.00140.

[3] N. Aslam, S. Srivastava and M. M. Gore, "Evaluating DDoS Detection and Mitigation in SDN at Various Attack Rates," 2024 First International Conference on Pioneering Developments in Computer Science & Digital Technologies (IC2SDT), Delhi, India, 2024, pp. 569-574, doi: 10.1109/IC2SDT62152.2024.10696232.

[4] S. F. Amin, A. B. Jadhav, P. V. Hatkar and M. M. Bachate, "A Lightweight RF-XGBoost-LinearSVM Ensemble for High-Performance DDoS Attack Detection Using the CICDDoS2019 Dataset," 2025 IEEE International Conference on Blockchain and Distributed Systems Security (ICBDS), Kolhapur, India, 2025, pp. 1-6, doi: 10.1109/ICBDS67396.2025.11378141.

[5] S. S. Mohapatra, S. Parija, D. Muduli, S. K. Sharma and R. R. Kumar, "Improved DDoS Attack Detection Through PCA-Enhanced Ensemble Machine Learning Approach," 2025 International Conference on Intelligent and Cloud Computing (ICoICC), Bhubaneswar, India, 2025, pp. 1-5, doi: 10.1109/ICoICC64033.2025.11051373.

[6] B. S. Khassawneh, F. Alanazi, A. Alsarhan, G. M. Jaradat, M. Aljamal and S. A. Alshammari, "Integrating One-Class Anomaly Detection and Ensemble Learning for Robust and Efficient IDS Design," 2025 37th International Conference on Microelectronics (ICM), Cairo, Egypt, 2025, pp. 1-6, doi: 10.1109/ICM66518.2025.11322503.

[7] S. W. Nourildean, W. K. Ahmed, A. A. Sabri and M. Aziz, "Intrusion Detection System Based Ensemble Machine Learning Model in AIoT Against Cyber Attacks," 2025 Advances in Science and Engineering Technology International Conferences (ASET), Dubai, United Arab Emirates, 2025, pp. 1-7, doi: 10.1109/ASET66891.2025.11428129.

[8] S. Chunduru and N. M. Rao Purimetla, "Hybrid Framework for Detecting Multi-Class DDoS Attacks Using Probabilistic and Boosted Learning Methods," 2025 IEEE 1st International Conference on Recent Trends in Computing and Smart Mobility (RCSM), Bhopal, India, 2025, pp. 1-6, doi: 10.1109/RCSM67767.2025.11508067.

[9] K. J. Handoyo, N. W. Hariyanto, B. S. Edipurtta, B. A. Jabar and J. V. Moniaga, "Utilizing AI-Powered Anomaly Detection to Counter DDoS Attacks: A Comparative Study of Random Forest and XGBoost Algorithm in Network Protection," 2025 7th International Conference on Cybernetics and Intelligent System (ICORIS), Mataram, Indonesia, 2025, pp. 1-6, doi: 10.1109/ICORIS67789.2025.11296019.

[10] U. Rajput, U. Roy, S. R. Chowdhury, V. Agarwal, A. U. Khan and D. K. Behera, "Random Forest-Based LR-DDoS Detection in Cloud: Enhanced Preprocessing and Comparative Evaluation," 2026 International Conference on Intelligent Computing, Networks, and Security (IC-ICNS), Bhubaneswar, India, 2026, pp. 1-7, doi: 10.1109/IC-ICNS68863.2026.11537636.

[11] J. Zhang, M. Zulkernine and A. Haque, "Random-Forests-Based Network Intrusion Detection Systems," in IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews), vol. 38, no. 5, pp. 649-659, Sept. 2008, doi: 10.1109/TSMCC.2008.923876.

[12] M. Kaur and S. Chouhan, "Enhanced DDoS Attack Detection Using Hybrid Machine Learning Approaches," 2025 7th International Symposium on Advanced Electrical and Communication Technologies (ISAECT), Mohali, Punjab, India, 2025, pp. 1-5, doi: 10.1109/ISAECT68904.2025.11318718.

[13] M. Dandotiya and R. R. Singh Makwana, "Improving Network Security with Hybrid Model for DDoS Attack Detection," 2024 IEEE International Conference on Blockchain and Distributed Systems Security (ICBDS), Pune, India, 2024, pp. 1-8, doi: 10.1109/ICBDS61829.2024.10837036.

[14] K. Kishor, P. Singh, R. J. Dixit, T. Jain, A. Sharma and K. Chaudhary, "Real-Time Flow-Based DDoS Detection and Adaptive Mitigation Using Machine Learning in Enterprise Networks," 2025 IEEE 7th International Conference on Computing, Communication and Automation (ICCCA), Greater Noida, India, 2025, pp. 1-7, doi: 10.1109/ICCCA66364.2025.11325275.

[15] X. A. Presskila, S. B. Setty and N. V. Krishna, "An Intelligent Machine Learning Framework for Ddos Attack Detection and Prediction," 2026 3rd International Conference on Research Methodologies in Knowledge Management, Artificial Intelligence and Telecommunication Engineering (RMKMATE), Chennai, India, 2026, pp. 1-5, doi: 10.1109/RMKMATE69073.2026.11519024.

[16] M. R. O. Aswath Prakash, R. Gayathri and M. Braveen, "A Secure Intrusion Detection System Against DDoS Attacks Using Machine Learning," 2026 International Conference on Data Science, Agents and Artificial Intelligence (ICDSAAI), Chennai, India, 2026, pp. 1-5, doi: 10.1109/ICDSAAI69492.2026.11505160.

[17] A. S. Liz. D.R. et al., "Adaptive Real-Time Meta-Learning Framework with Hybrid AI Classifiers for Live Detection and Mitigation of DDoS Attacks," 2025 International Conference on Innovations and Emerging Technologies In AI & Communication Systems (IETACS), Mohali, India, 2025, pp. 1637-1643, doi: 10.1109/IETACS68750.2025.11385481

[18] N. Khan, R. Alharthi, F. Alzahrani, M. Benmalek and A. Derhab, "DDoS Attack Detection in Cloud Systems Using a Hybrid Stacked Ensemble," 2026 6th Middle East and North Africa Communications Conference (MENACOMM), Hammamet, Tunisia, 2026, pp. 1-7, doi: 10.1109/MENACOMM69507.2026.11532804.

[19] R. B. Mallela, S. O, E. A. Goud, A. Rapaka, P. R. Sruthi and S. Naga Yasaswi, "Optimizing IoT DDoS Detection with Hybrid Feature Selection and Ensemble Learning," 2025 3rd International Conference on Sustainable Computing and Data Communication Systems (ICSCDS), Erode, India, 2025, pp. 1004-1008, doi: 10.1109/ICSCDS65426.2025.11166901.

[20] S. D. Gonuguntla, S. JayaPrakash and R. H. Sai, "A Hybrid Ensemble Model for Network Intrusion Detection," 2025 12th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO), Noida NCR, India, 2025, pp. 1-4, doi: 10.1109/ICRITO66076.2025.11241516.